

Compound Renewal Processes: Distribution Analysis and Properties

Roberto Magno Mazzotta

July 2025

Problem definition

Consider a sequence of i.i.d. positive and absolutely continuous random variables $\{J_i\}_{i=1}^{\infty}$ with the meaning of interevent times, define the following processes

$$T_n = \sum_{i=1}^n J_i, \quad T_0 = 0,$$

and

$$N(t) = \max\{n : T_n \leq t\}.$$

Now, consider a sequence of i.i.d absolutely continuous random variables $\{X_i\}_{i=1}^{\infty}$ with the meaning of jumps and define the following process

$$Z_n = \sum_{i=1}^n X_i, \quad Z_0 = 0.$$

Finally consider this process

$$Z(t) = Z_{N(t)} = \sum_{i=1}^{N(t)} X_i, \quad Z(0) = Z_0 = 0.$$

1. From the knowledge of the cumulative distribution function, $F_{J_1}(u)$ and $F_{X_1}(u)$ and the respective probability density functions derive the cumulative distribution functions and the probability density functions of T_n , $N(t)$, Z_n , and $Z(t)$.
2. Explicitly write the finite dimensional distributions of the process $Z(t)$.

3. Prove the existence of the process $Z(t)$. What can you say on its sample paths?
4. Find the conditions under which $Z(t)$ is a Markov process.
5. Find the conditions under which $Z(t)$ is a martingale.

The work is organized as follows: Section 1 derives the distributions of the constituent processes. Section 2 analyzes the finite-dimensional distributions of $Z(t)$. Section 3 proves the existence of the process and analyzes its sample path properties. Sections 4 and 5 establish the conditions for the Markov and martingale properties, respectively.

1 Distribution Analysis

1.1 Distribution of T_n

T_n represents the time of the n -th event and is defined as the sum of n i.i.d. random variables J_i . The distribution of T_n depends directly on the distribution of the interevent times J_i .

For $n = 1$, $T_1 = J_1$, so:

$$\begin{aligned} F_{T_1}(t) &= F_{J_1}(t) \\ f_{T_1}(t) &= f_{J_1}(t) \end{aligned}$$

For $n = 2$, $T_2 = J_1 + J_2$, and since J_1 and J_2 are independent:

$$\begin{aligned} F_{T_2}(t) &= \int_0^t F_{J_1}(t-s) f_{J_1}(s) ds \\ f_{T_2}(t) &= \int_0^t f_{J_1}(t-s) f_{J_1}(s) ds = (f_{J_1} * f_{J_1})(t) \end{aligned}$$

where the domain of integration is $[0, t]$ since the interevent times J_i are positive.

By induction, for any $n \geq 1$:

$$\begin{aligned} F_{T_n}(t) &= F_{J_1}^{*n}(t) \\ f_{T_n}(t) &= f_{J_1}^{*n}(t) \end{aligned}$$

where $F_{J_1}^{*n}$ and $f_{J_1}^{*n}$ denote the n -fold convolution of F_{J_1} and f_{J_1} , respectively feller.

For specific distributions of J_i , we can determine the exact distribution of T_n :

1. If $J_i \sim \text{Exponential}(\lambda)$, then $T_n \sim \text{Gamma}(n, \lambda)$ with density:

$$f_{T_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{\Gamma(n)}, \quad t > 0$$

2. If $J_i \sim \text{Gamma}(\alpha, \beta)$, then $T_n \sim \text{Gamma}(n\alpha, \beta)$ with density:

$$f_{T_n}(t) = \frac{\beta^{n\alpha} t^{n\alpha-1} e^{-\beta t}}{\Gamma(n\alpha)}, \quad t > 0$$

3. If $J_i \sim \text{Uniform}(0, b)$, then T_n follows an n -fold convolution of uniform distributions, with a piecewise polynomial density.

For general distributions, we can use the Laplace transform method. If $\mathcal{L}[f_{J_1}](s) = \int_0^\infty e^{-st} f_{J_1}(t) dt$ is the Laplace transform of $f_{J_1}(t)$, then:

$$\mathcal{L}[f_{T_n}](s) = [\mathcal{L}[f_{J_1}](s)]^n$$

The PDF $f_{T_n}(t)$ can be obtained by the inverse Laplace transform:

$$f_{T_n}(t) = \mathcal{L}^{-1}\{[\mathcal{L}[f_{J_1}](s)]^n\}(t)$$

This inversion is guaranteed to exist when $f_{J_1}(t)$ has finite moments tjms2003first .

1.2 Distribution of $N(t)$

$N(t)$ counts the number of events up to time t . Its distribution is determined by the distribution of T_n , which in turn depends on the distribution of J_i .

For any non-negative integer n :

$$\begin{aligned} \{N(t) \geq n\} &\Leftrightarrow \{T_n \leq t\} \\ \{N(t) = n\} &\Leftrightarrow \{T_n \leq t < T_{n+1}\} \end{aligned}$$

Therefore:

$$\begin{aligned} P(N(t) \geq n) &= P(T_n \leq t) = F_{T_n}(t) \\ P(N(t) = n) &= P(T_n \leq t < T_{n+1}) \\ &= P(T_n \leq t) - P(T_{n+1} \leq t) \\ &= F_{T_n}(t) - F_{T_{n+1}}(t) \end{aligned}$$

Let's verify that these probabilities sum to 1, which is a necessary condition for a valid probability mass function:

$$\begin{aligned}\sum_{n=0}^{\infty} P(N(t) = n) &= \sum_{n=0}^{\infty} [F_{T_n}(t) - F_{T_{n+1}}(t)] \\ &= F_{T_0}(t) - \lim_{n \rightarrow \infty} F_{T_{n+1}}(t)\end{aligned}$$

Since $T_0 = 0$, we have $F_{T_0}(t) = 1$ for $t \geq 0$. Also, since $T_n \rightarrow \infty$ as $n \rightarrow \infty$ (given that $J_i > 0$), we have $\lim_{n \rightarrow \infty} F_{T_{n+1}}(t) = 0$ for any fixed t . Therefore:

$$\sum_{n=0}^{\infty} P(N(t) = n) = 1 - 0 = 1$$

This confirms that our probabilities form a valid PMF.

For specific distributions of J_i , we can determine explicit expressions for the PMF of $N(t)$:

1. If $J_i \sim \text{Exponential}(\lambda)$, then $N(t)$ is a Poisson process with rate λ :

$$P(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad n = 0, 1, 2, \dots$$

2. If $J_i \sim \text{Gamma}(\alpha, \beta)$, then the PMF involves differences of incomplete gamma functions:

$$P(N(t) = n) = \frac{\Gamma(n\alpha, \beta t)}{\Gamma(n\alpha)} - \frac{\Gamma((n+1)\alpha, \beta t)}{\Gamma((n+1)\alpha)}$$

where $\Gamma(a, x) = \int_x^{\infty} u^{a-1} e^{-u} du$ is the upper incomplete gamma function.

The cumulative distribution function of $N(t)$ is:

$$\begin{aligned}F_{N(t)}(n) &= P(N(t) \leq n) \\ &= P(N(t) < n+1) \\ &= 1 - P(N(t) \geq n+1) \\ &= 1 - F_{T_{n+1}}(t)\end{aligned}$$

Since $N(t)$ is a discrete random variable, it does not have a probability density function in the usual sense, but rather a probability mass function (PMF):

$$P(N(t) = n) = F_{T_n}(t) - F_{T_{n+1}}(t)$$

1.3 Distribution of Z_n

Z_n represents the cumulative sum of the first n jumps. Similar to T_n , the distribution of Z_n can be derived using convolution operations, since Z_n is the sum of n i.i.d. random variables X_i .

For $n = 1$, $Z_1 = X_1$, so:

$$\begin{aligned} F_{Z_1}(z) &= F_{X_1}(z) \\ f_{Z_1}(z) &= f_{X_1}(z) \end{aligned}$$

By the same reasoning as for T_n , for any $n \geq 1$:

$$\begin{aligned} F_{Z_n}(z) &= F_{X_1}^{*n}(z) \\ f_{Z_n}(z) &= f_{X_1}^{*n}(z) \end{aligned}$$

For specific distributions of X_i , we can determine the exact distribution of Z_n :

1. If $X_i \sim \text{Normal}(\mu, \sigma^2)$, then $Z_n \sim \text{Normal}(n\mu, n\sigma^2)$ with density:

$$f_{Z_n}(z) = \frac{1}{\sqrt{2\pi n\sigma^2}} \exp\left(-\frac{(z - n\mu)^2}{2n\sigma^2}\right)$$

2. If $X_i \sim \text{Exponential}(\lambda)$, then $Z_n \sim \text{Gamma}(n, \lambda)$ with density:

$$f_{Z_n}(z) = \frac{\lambda^n z^{n-1} e^{-\lambda z}}{\Gamma(n)}, \quad z > 0$$

Using characteristic functions, if $\phi_{X_1}(s) = \mathbb{E}[e^{isX_1}]$ is the characteristic function of X_1 , then:

$$\phi_{Z_n}(s) = [\phi_{X_1}(s)]^n$$

The PDF can be recovered through the inverse Fourier transform:

$$f_{Z_n}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-isz} [\phi_{X_1}(s)]^n ds$$

This inversion exists under mild conditions, typically satisfied when X_1 has a finite variance, ensuring that $\phi_{X_1}(s)$ is square-integrable karlin.

1.4 Distribution of $Z(t)$

$Z(t) = Z_{N(t)}$ is the compound renewal process, representing the accumulated value up to time t . Its distribution depends on both the distributions of J_i (through $N(t)$) and X_i (through Z_n).

The cumulative distribution function of $Z(t)$ is:

$$\begin{aligned} F_{Z(t)}(z) &= P(Z(t) \leq z) \\ &= \sum_{n=0}^{\infty} P(Z(t) \leq z \mid N(t) = n) P(N(t) = n) \end{aligned}$$

Given $N(t) = n$, we have $Z(t) = Z_n$. Therefore:

$$\begin{aligned} F_{Z(t)}(z) &= \sum_{n=0}^{\infty} P(Z_n \leq z) P(N(t) = n) \\ &= \sum_{n=0}^{\infty} F_{Z_n}(z) [F_{T_n}(t) - F_{T_{n+1}}(t)] \\ &= \sum_{n=0}^{\infty} F_{X_1}^{*n}(z) [F_{J_1}^{*n}(t) - F_{J_1}^{*(n+1)}(t)] \end{aligned}$$

This expression clearly shows how $Z(t)$ depends on both the jump distribution X_1 and the interevent time distribution J_1 .

For the PDF of $Z(t)$, we differentiate the CDF with respect to z :

$$\begin{aligned} f_{Z(t)}(z) &= \frac{d}{dz} F_{Z(t)}(z) \\ &= \sum_{n=0}^{\infty} f_{Z_n}(z) [F_{T_n}(t) - F_{T_{n+1}}(t)] \\ &= \sum_{n=0}^{\infty} f_{X_1}^{*n}(z) [F_{J_1}^{*n}(t) - F_{J_1}^{*(n+1)}(t)] \end{aligned}$$

For specific distributions of both J_i and X_i , we can derive more explicit expressions. For example, in the important special case of a compound Poisson process where $J_i \sim \text{Exponential}(\lambda)$ and $X_i \sim F_X$:

$$\begin{aligned} f_{Z(t)}(z) &= \sum_{n=0}^{\infty} f_{X_1}^{*n}(z) \cdot e^{-\lambda t} \frac{(\lambda t)^n}{n!} \\ &= e^{-\lambda t} \delta(z) + e^{-\lambda t} \sum_{n=1}^{\infty} f_{X_1}^{*n}(z) \frac{(\lambda t)^n}{n!} \end{aligned}$$

where $\delta(z)$ is the Dirac delta function, accounting for the case $n = 0$.

The characteristic function of $Z(t)$ for a compound Poisson process has the particularly elegant form:

$$\phi_{Z(t)}(s) = e^{\lambda t(\phi_{X_1}(s)-1)}$$

as derived in [7].

2 Finite Dimensional Distributions of the Process $Z(t)$

In this section I derive the finite dimensional distributions of the compound renewal process $Z(t)$.

2.1 Definition of Finite Dimensional Distributions

The finite dimensional distributions of a stochastic process $Z(t)$ characterize the joint distribution of the process at any finite collection of time points. Specifically, for any $n \geq 1$ and any set of time points $0 \leq t_1 < t_2 < \dots < t_n$, the finite dimensional distribution describes the joint probability:

$$P(Z(t_1) \leq z_1, Z(t_2) \leq z_2, \dots, Z(t_n) \leq z_n)$$

where $z_1, z_2, \dots, z_n$ are real numbers.

2.2 Derivation for Compound Renewal Processes

To derive the finite dimensional distributions of $Z(t)$, we use the fact that the process is fully determined by the counting process $N(t)$ and the sequence of jumps $\{X_i\}_{i=1}^{\infty}$.

Let us consider the case of $n = 2$ time points $t_1 < t_2$ to illustrate the approach, then generalize to arbitrary n .

2.2.1 Two-Dimensional Distribution

For $t_1 < t_2$, we have:

$$Z(t_1) = \sum_{i=1}^{N(t_1)} X_i$$

$$Z(t_2) = \sum_{i=1}^{N(t_2)} X_i = Z(t_1) + \sum_{i=N(t_1)+1}^{N(t_2)} X_i$$

To find the joint distribution $P(Z(t_1) \leq z_1, Z(t_2) \leq z_2)$, we condition on the values of the counting process:

$$P(Z(t_1) \leq z_1, Z(t_2) \leq z_2)$$

$$= \sum_{k=0}^{\infty} \sum_{m=k}^{\infty} P(Z(t_1) \leq z_1, Z(t_2) \leq z_2 \mid N(t_1) = k, N(t_2) = m) \cdot P(N(t_1) = k, N(t_2) = m)$$

Note that $N(t_2) \geq N(t_1)$ since the counting process is non-decreasing. Now, conditional on $N(t_1) = k$ and $N(t_2) = m$, it follows:

$$Z(t_1) = \sum_{i=1}^k X_i = Z_k$$

$$Z(t_2) = \sum_{i=1}^m X_i = Z_m$$

Thus:

$$P(Z(t_1) \leq z_1, Z(t_2) \leq z_2 \mid N(t_1) = k, N(t_2) = m)$$

$$= P(Z_k \leq z_1, Z_m \leq z_2 \mid N(t_1) = k, N(t_2) = m)$$

Since $\{X_i\}_{i=1}^{\infty}$ are independent of the counting process $N(t)$, we have:

$$P(Z_k \leq z_1, Z_m \leq z_2 \mid N(t_1) = k, N(t_2) = m) = P(Z_k \leq z_1, Z_m \leq z_2)$$

We can decompose Z_m as:

$$Z_m = Z_k + (Z_m - Z_k) = Z_k + \sum_{i=k+1}^m X_i$$

Since Z_k and $\sum_{i=k+1}^m X_i$ are independent (due to the independence of the X_i 's), it follows:

$$\begin{aligned}
P(Z_k \leq z_1, Z_m \leq z_2) &= P(Z_k \leq z_1, Z_k + \sum_{i=k+1}^m X_i \leq z_2) \\
&= \int_{-\infty}^{z_1} P(Z_k + \sum_{i=k+1}^m X_i \leq z_2 \mid Z_k = y) \cdot f_{Z_k}(y) dy \\
&= \int_{-\infty}^{z_1} P(\sum_{i=k+1}^m X_i \leq z_2 - y) \cdot f_{Z_k}(y) dy \\
&= \int_{-\infty}^{z_1} F_{Z_{m-k}}(z_2 - y) \cdot f_{Z_k}(y) dy
\end{aligned}$$

where $F_{Z_{m-k}}$ is the CDF of the sum of $(m - k)$ i.i.d. random variables X_i .

Therefore:

$$\begin{aligned}
P(Z(t_1) \leq z_1, Z(t_2) \leq z_2) \\
= \sum_{k=0}^{\infty} \sum_{m=k}^{\infty} \left[\int_{-\infty}^{z_1} F_{Z_{m-k}}(z_2 - y) \cdot f_{Z_k}(y) dy \right] \cdot P(N(t_1) = k, N(t_2) = m)
\end{aligned}$$

The joint probability $P(N(t_1) = k, N(t_2) = m)$ depends on the specific properties of the renewal process. For a general renewal process, this joint probability can be expressed as:

$$P(N(t_1) = k, N(t_2) = m) = P(N(t_1) = k) \cdot P(N(t_2) - N(t_1) = m - k \mid N(t_1) = k)$$

For a general renewal process, $P(N(t_2) - N(t_1) = m - k \mid N(t_1) = k)$ depends on the age of the renewal process at time t_1 , which complicates the analysis. However, in the special case where the interevent times $\{J_i\}$ are exponentially distributed (resulting in a Poisson process), the process has the memoryless property, and we have:

$$P(N(t_2) - N(t_1) = m - k \mid N(t_1) = k) = P(N(t_2 - t_1) = m - k)$$

This significantly simplifies the joint distribution in the Poisson case.

2.2.2 General Finite Dimensional Distribution

Now, we can generalize this approach to derive the n -dimensional distribution for any $n \geq 1$ and times $0 \leq t_1 < t_2 < \dots < t_n$.

We have:

$$\begin{aligned}
& P(Z(t_1) \leq z_1, Z(t_2) \leq z_2, \dots, Z(t_n) \leq z_n) \\
&= \sum_{k_1=0}^{\infty} \sum_{k_2=k_1}^{\infty} \cdots \sum_{k_n=k_{n-1}}^{\infty} P(Z(t_1) \leq z_1, \dots, Z(t_n) \leq z_n \mid N(t_1) = k_1, \dots, N(t_n) = k_n) \\
&\quad \times P(N(t_1) = k_1, N(t_2) = k_2, \dots, N(t_n) = k_n)
\end{aligned}$$

Given $N(t_j) = k_j$ for $j = 1, 2, \dots, n$, we have $Z(t_j) = Z_{k_j}$. The conditional probability becomes:

$$\begin{aligned}
& P(Z(t_1) \leq z_1, \dots, Z(t_n) \leq z_n \mid N(t_1) = k_1, \dots, N(t_n) = k_n) \\
&= P(Z_{k_1} \leq z_1, Z_{k_2} \leq z_2, \dots, Z_{k_n} \leq z_n)
\end{aligned}$$

Decomposing each Z_{k_j} in terms of increments:

$$\begin{aligned}
Z_{k_1} &= \sum_{i=1}^{k_1} X_i \\
Z_{k_2} &= Z_{k_1} + \sum_{i=k_1+1}^{k_2} X_i \\
&\vdots \\
Z_{k_n} &= Z_{k_{n-1}} + \sum_{i=k_{n-1}+1}^{k_n} X_i
\end{aligned}$$

Let's define the increments $Y_j = \sum_{i=k_{j-1}+1}^{k_j} X_i$ for $j = 1, 2, \dots, n$ (with $k_0 = 0$). Then:

$$\begin{aligned}
Z_{k_1} &= Y_1 \\
Z_{k_2} &= Y_1 + Y_2 \\
&\vdots \\
Z_{k_n} &= Y_1 + Y_2 + \cdots + Y_n
\end{aligned}$$

Since the Y_j 's are independent (as they involve disjoint sets of the i.i.d. random variables X_i), we can write:

$$\begin{aligned}
& P(Z_{k_1} \leq z_1, Z_{k_2} \leq z_2, \dots, Z_{k_n} \leq z_n) \\
&= P(Y_1 \leq z_1, Y_1 + Y_2 \leq z_2, \dots, Y_1 + Y_2 + \cdots + Y_n \leq z_n)
\end{aligned}$$

This probability can be evaluated by iterated integrals:

$$\begin{aligned} & P(Y_1 \leq z_1, Y_1 + Y_2 \leq z_2, \dots, Y_1 + Y_2 + \dots + Y_n \leq z_n) \\ &= \int_{-\infty}^{z_1} \int_{-\infty}^{z_2 - y_1} \dots \int_{-\infty}^{z_n - y_1 - \dots - y_{n-1}} f_{Y_1}(y_1) f_{Y_2}(y_2) \dots f_{Y_n}(y_n) dy_n \dots dy_2 dy_1 \end{aligned}$$

where $f_{Y_j}(y_j)$ is the PDF of $Y_j = \sum_{i=k_{j-1}+1}^{k_j} X_i$, which follows the distribution of the sum of $(k_j - k_{j-1})$ i.i.d. random variables X_i .

Therefore, the complete finite dimensional distribution is:

$$\begin{aligned} & P(Z(t_1) \leq z_1, Z(t_2) \leq z_2, \dots, Z(t_n) \leq z_n) \\ &= \sum_{k_1=0}^{\infty} \sum_{k_2=k_1}^{\infty} \dots \sum_{k_n=k_{n-1}}^{\infty} \left[\int_{-\infty}^{z_1} \dots \int_{-\infty}^{z_n - y_1 - \dots - y_{n-1}} \prod_{j=1}^n f_{Y_j}(y_j) dy_n \dots dy_1 \right] \\ & \quad \times P(N(t_1) = k_1, N(t_2) = k_2, \dots, N(t_n) = k_n) \end{aligned}$$

The joint probability $P(N(t_1) = k_1, N(t_2) = k_2, \dots, N(t_n) = k_n)$ can be factorized as:

$$\begin{aligned} & P(N(t_1) = k_1, N(t_2) = k_2, \dots, N(t_n) = k_n) \\ &= P(N(t_1) = k_1) \cdot P(N(t_2) - N(t_1) = k_2 - k_1 \mid N(t_1) = k_1) \\ & \quad \times P(N(t_3) - N(t_2) = k_3 - k_2 \mid N(t_1) = k_1, N(t_2) = k_2) \times \dots \end{aligned}$$

For a general renewal process, these conditional probabilities depend on the age of the process, making the expression quite complex. However, as noted earlier, for a Poisson process, the memoryless property leads to significant simplification:

$$P(N(t_1) = k_1, N(t_2) = k_2, \dots, N(t_n) = k_n) = P(N(t_1) = k_1) \cdot \prod_{j=2}^n P(N(t_j - t_{j-1}) = k_j - k_{j-1})$$

2.3 Special Case: Compound Poisson Process

When the interevent times $\{J_i\}$ are exponentially distributed with parameter λ , the counting process $N(t)$ is a Poisson process with rate λ . In this case, the compound process $Z(t)$ is called a compound Poisson process, and its finite dimensional distributions take a more tractable form.

For a compound Poisson process, we have:

$$P(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$$

And the independent increments property gives:

$$P(N(t_j) - N(t_{j-1}) = k_j - k_{j-1}) = e^{-\lambda(t_j - t_{j-1})} \frac{[\lambda(t_j - t_{j-1})]^{k_j - k_{j-1}}}{(k_j - k_{j-1})!}$$

Thus, the joint probability of the counting process becomes:

$$P(N(t_1) = k_1, \dots, N(t_n) = k_n) = e^{-\lambda t_n} \prod_{j=1}^n \frac{[\lambda(t_j - t_{j-1})]^{k_j - k_{j-1}}}{(k_j - k_{j-1})!}$$

where $t_0 = 0$ and $k_0 = 0$.

Substituting this into our general formula, we get the finite dimensional distributions for a compound Poisson process:

$$\begin{aligned} & P(Z(t_1) \leq z_1, \dots, Z(t_n) \leq z_n) \\ &= \sum_{k_1=0}^{\infty} \sum_{k_2=k_1}^{\infty} \dots \sum_{k_n=k_{n-1}}^{\infty} \left[\int_{-\infty}^{z_1} \dots \int_{-\infty}^{z_n - y_1 - \dots - y_{n-1}} \prod_{j=1}^n f_{Y_j}(y_j) dy_n \dots dy_1 \right] \\ & \times e^{-\lambda t_n} \prod_{j=1}^n \frac{[\lambda(t_j - t_{j-1})]^{k_j - k_{j-1}}}{(k_j - k_{j-1})!} \end{aligned}$$

2.4 Characteristic Function Approach

An alternative and often more tractable approach to derive the finite dimensional distributions is through characteristic functions. Let $\phi_{Z(t)}(s) = \mathbb{E}[e^{isZ(t)}]$ be the characteristic function of $Z(t)$.

The joint characteristic function of $(Z(t_1), Z(t_2), \dots, Z(t_n))$ is defined as:

$$\phi_{Z(t_1), Z(t_2), \dots, Z(t_n)}(s_1, s_2, \dots, s_n) = \mathbb{E}[e^{i(s_1 Z(t_1) + s_2 Z(t_2) + \dots + s_n Z(t_n))}]$$

For the compound renewal process, we can derive this by conditioning on the values of the counting process:

$$\begin{aligned} & \phi_{Z(t_1), Z(t_2), \dots, Z(t_n)}(s_1, s_2, \dots, s_n) \\ &= \sum_{k_1=0}^{\infty} \sum_{k_2=k_1}^{\infty} \dots \sum_{k_n=k_{n-1}}^{\infty} \mathbb{E}[e^{i(s_1 Z(t_1) + s_2 Z(t_2) + \dots + s_n Z(t_n))} \mid N(t_1) = k_1, \dots, N(t_n) = k_n] \\ & \times P(N(t_1) = k_1, N(t_2) = k_2, \dots, N(t_n) = k_n) \end{aligned}$$

Given $N(t_j) = k_j$, we have $Z(t_j) = Z_{k_j}$. Using the decomposition into increments Y_j as before:

$$\begin{aligned} & \mathbb{E}[e^{i(s_1 Z(t_1) + s_2 Z(t_2) + \dots + s_n Z(t_n))} \mid N(t_1) = k_1, \dots, N(t_n) = k_n] \\ &= \mathbb{E}[e^{i(s_1 Y_1 + s_2 (Y_1 + Y_2) + \dots + s_n (Y_1 + Y_2 + \dots + Y_n))}] \\ &= \mathbb{E}[e^{i((s_1 + s_2 + \dots + s_n) Y_1 + (s_2 + \dots + s_n) Y_2 + \dots + s_n Y_n)}] \end{aligned}$$

Since the Y_j 's are independent, this becomes:

$$\begin{aligned} & \mathbb{E}[e^{i((s_1 + s_2 + \dots + s_n) Y_1 + (s_2 + \dots + s_n) Y_2 + \dots + s_n Y_n)}] \\ &= \mathbb{E}[e^{i(s_1 + s_2 + \dots + s_n) Y_1}] \cdot \mathbb{E}[e^{i(s_2 + \dots + s_n) Y_2}] \cdot \dots \cdot \mathbb{E}[e^{i s_n Y_n}] \\ &= \phi_{Y_1}(s_1 + s_2 + \dots + s_n) \cdot \phi_{Y_2}(s_2 + \dots + s_n) \cdot \dots \cdot \phi_{Y_n}(s_n) \end{aligned}$$

where $\phi_{Y_j}(s)$ is the characteristic function of $Y_j = \sum_{i=k_{j-1}+1}^{k_j} X_i$.

Given that Y_j is the sum of $(k_j - k_{j-1})$ i.i.d. random variables X_i , we have:

$$\phi_{Y_j}(s) = [\phi_X(s)]^{k_j - k_{j-1}}$$

where $\phi_X(s)$ is the characteristic function of a single jump X_i .

Therefore:

$$\begin{aligned} & \mathbb{E}[e^{i(s_1 Z(t_1) + s_2 Z(t_2) + \dots + s_n Z(t_n))} \mid N(t_1) = k_1, \dots, N(t_n) = k_n] \\ &= [\phi_X(s_1 + s_2 + \dots + s_n)]^{k_1} \cdot [\phi_X(s_2 + \dots + s_n)]^{k_2 - k_1} \cdot \dots \cdot [\phi_X(s_n)]^{k_n - k_{n-1}} \end{aligned}$$

The full joint characteristic function is then:

$$\begin{aligned} & \phi_{Z(t_1), Z(t_2), \dots, Z(t_n)}(s_1, s_2, \dots, s_n) \\ &= \sum_{k_1=0}^{\infty} \sum_{k_2=k_1}^{\infty} \dots \sum_{k_n=k_{n-1}}^{\infty} [\phi_X(s_1 + s_2 + \dots + s_n)]^{k_1} \cdot [\phi_X(s_2 + \dots + s_n)]^{k_2 - k_1} \cdot \dots \cdot [\phi_X(s_n)]^{k_n - k_{n-1}} \\ &\quad \times P(N(t_1) = k_1, N(t_2) = k_2, \dots, N(t_n) = k_n) \end{aligned}$$

For the special case of a compound Poisson process, this further simplifies to:

$$\begin{aligned} & \phi_{Z(t_1), Z(t_2), \dots, Z(t_n)}(s_1, s_2, \dots, s_n) \\ &= \exp \left\{ \lambda t_1 (\phi_X(s_1 + s_2 + \dots + s_n) - 1) + \lambda (t_2 - t_1) (\phi_X(s_2 + \dots + s_n) - 1) \right. \\ &\quad \left. + \dots + \lambda (t_n - t_{n-1}) (\phi_X(s_n) - 1) \right\} \end{aligned}$$

This expression for the joint characteristic function completely characterizes the finite dimensional distributions of a compound Poisson process

3 Existence of the Process $Z(t)$

In this section I prove the existence of the compound renewal process $Z(t)$ and analyze its sample path properties.

3.1 Constructive Proof of Existence

The existence of the process $Z(t)$ can be established using a constructive approach. We first establish the existence of the component processes and then show how they combine to form a well-defined stochastic process.

3.1.1 Existence of the Renewal Counting Process $N(t)$

The renewal counting process $N(t)$ is defined as $N(t) = \max\{n : T_n \leq t\}$, where $T_n = \sum_{i=1}^n J_i$ are the renewal times.

- For the sequence $\{J_i\}_{i=1}^\infty$ of i.i.d. positive random variables, the existence is guaranteed by the standard probability theory of i.i.d. sequences on a probability space $(\Omega, \mathcal{F}, P)$.
- Given the sequence $\{J_i\}_{i=1}^\infty$, the partial sums $T_n = \sum_{i=1}^n J_i$ are well-defined random variables for each $n \geq 1$, with $T_0 = 0$.
- Since $J_i > 0$ for all i , the sequence $\{T_n\}_{n=0}^\infty$ is strictly increasing with probability 1, i.e., $T_0 < T_1 < T_2 < \dots$ almost surely.
- For any $t \geq 0$, define $N(t) = \max\{n : T_n \leq t\} = \sum_{n=0}^\infty \mathbf{1}_{\{T_n \leq t\}}$. This is well-defined because for any $t < \infty$, only finitely many terms in the sum are non-zero (since $T_n \rightarrow \infty$ as $n \rightarrow \infty$ with probability 1).

Therefore, $N(t)$ exists as a well-defined stochastic process on $[0, \infty)$.

3.1.2 Existence of the Compound Process $Z(t)$

Now, let us establish the existence of the compound process $Z(t)$.

- The sequence $\{X_i\}_{i=1}^\infty$ of i.i.d. absolutely continuous random variables exists on the probability space $(\Omega, \mathcal{F}, P)$.

- For each $n \geq 1$, the partial sum $Z_n = \sum_{i=1}^n X_i$ is a well-defined random variable, with $Z_0 = 0$.
- Given the renewal counting process $N(t)$, the compound process $Z(t) = Z_{N(t)} = \sum_{i=1}^{N(t)} X_i$ is defined as the composition of Z_n and $N(t)$.
- For any fixed $t \geq 0$, $N(t)$ is a non-negative integer-valued random variable. Since Z_n is defined for all non-negative integers n , the composition $Z_{N(t)}$ is well-defined.
- This defines $Z(t)$ for all $t \geq 0$ as a stochastic process.

3.2 Kolmogorov's Extension Theorem Approach

An alternative approach to establishing the existence of $Z(t)$ is through Kolmogorov's Extension Theorem, which provides conditions under which a consistent family of finite-dimensional distributions can be extended to a full stochastic process.

- In Section 2, we derived the finite-dimensional distributions of $Z(t)$.
- These finite-dimensional distributions satisfy the Kolmogorov consistency conditions:
 1. Symmetry: The distributions are invariant under permutations of indices that preserve the ordering of time points.
 2. Marginalization: For any $m < n$ and indices $1 \leq i_1 < i_2 < \dots < i_m \leq n$, the distribution of $(Z(t_{i_1}), \dots, Z(t_{i_m}))$ can be obtained from the distribution of $(Z(t_1), \dots, Z(t_n))$ by marginalization.
- By Kolmogorov's Extension Theorem, there exists a stochastic process $\{Z(t), t \geq 0\}$ with these finite-dimensional distributions.

This establishes the existence of the compound renewal process $Z(t)$ through two different approaches: direct construction and the application of Kolmogorov's Extension Theorem.

3.3 Sample Path Properties of $Z(t)$

Having established the existence of the compound renewal process $Z(t)$, we now analyze its sample path properties.

3.3.1 Right-Continuity with Left Limits

The compound renewal process $Z(t)$ exhibits the càdlàg property (right-continuous with left limits).

- Right-continuity: For any $t \geq 0$, $\lim_{s \downarrow t} Z(s) = Z(t)$ almost surely.
- Left limits: For any $t > 0$, the limit $\lim_{s \uparrow t} Z(s)$ exists almost surely.

First, observe that the renewal counting process $N(t)$ is right-continuous with left limits:

1. Right-continuity: For any $t \geq 0$, if $N(t) = n$, then $T_n \leq t < T_{n+1}$. For any s such that $t < s < T_{n+1}$, we also have $N(s) = n$. Therefore, $\lim_{s \downarrow t} N(s) = N(t)$.
2. Left limits: For any $t > 0$, if $N(t) = n$ and $T_n < t$, then for any s such that $T_n < s < t$, we have $N(s) = n$, so $\lim_{s \uparrow t} N(s) = N(t)$. If $N(t) = n$ and $T_n = t$, then for any s such that $T_{n-1} < s < t$, we have $N(s) = n - 1$, so $\lim_{s \uparrow t} N(s) = n - 1 = N(t) - 1$.

Now, for the compound process $Z(t) = Z_{N(t)}$:

1. Right-continuity: Since $N(t)$ is right-continuous and Z_n is constant for fixed n , $Z(t) = Z_{N(t)}$ is right-continuous.
2. Left limits: For any $t > 0$, the left limit of $Z(t)$ is:

$$\begin{aligned} \lim_{s \uparrow t} Z(s) &= \lim_{s \uparrow t} Z_{N(s)} \\ &= Z_{\lim_{s \uparrow t} N(s)} \end{aligned}$$

If t is not a renewal time, then $\lim_{s \uparrow t} N(s) = N(t)$, so $\lim_{s \uparrow t} Z(s) = Z_{N(t)} = Z(t)$.

If t is a renewal time, say $t = T_n$ for some n , then $\lim_{s \uparrow t} N(s) = n - 1$, so $\lim_{s \uparrow t} Z(s) = Z_{n-1}$, which is different from $Z(t) = Z_n$.

3.3.2 Jump Discontinuities

The sample paths of $Z(t)$ have jump discontinuities at the renewal times $\{T_n\}_{n=1}^\infty$.

- At a renewal time $t = T_n$, the process $Z(t)$ jumps from Z_{n-1} to Z_n , with the jump size being X_n :

$$Z(T_n) - \lim_{s \uparrow T_n} Z(s) = Z_n - Z_{n-1} = X_n$$

- Between renewal times, the process $Z(t)$ remains constant. For $T_n < t < T_{n+1}$, we have $Z(t) = Z_n$.

3.3.3 Piecewise Constant Sample Paths

The sample paths of $Z(t)$ are piecewise constant functions with jumps at the renewal times.

- For any realization $\omega \in \Omega$, the sample path $t \mapsto Z(t, \omega)$ is a step function that changes values only at the renewal times $\{T_n(\omega)\}_{n=1}^\infty$.
- The function is constant on each interval $[T_n, T_{n+1})$, with value $Z_n(\omega)$.
- The jump sizes are given by the random variables $\{X_n\}_{n=1}^\infty$.

3.3.4 Measurability and Adaptedness

The compound renewal process $Z(t)$ possesses important measurability properties.

- $Z(t)$ is adapted to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$, where $\mathcal{F}_t = \sigma(N(s), X_1, X_2, \dots, X_{N(t)} : 0 \leq s \leq t)$ is the σ -algebra generated by the history of the process up to time t .
- $Z(t)$ is a càdlàg process, which implies that it is progressively measurable with respect to $\{\mathcal{F}_t\}_{t \geq 0}$.

3.3.5 Long-Term Behavior

Under mild conditions, the compound renewal process $Z(t)$ exhibits a linear growth behavior over long time scales.

- If $\mathbb{E}[J_1] < \infty$ and $\mathbb{E}[X_1] < \infty$, then by the Law of Large Numbers and the Elementary Renewal Theorem:

$$\lim_{t \rightarrow \infty} \frac{Z(t)}{t} = \frac{\mathbb{E}[X_1]}{\mathbb{E}[J_1]} \quad \text{almost surely}$$

- This means that the process grows approximately linearly with a slope given by the ratio of the expected jump size to the expected interevent time [2].

3.3.6 Special Case: Compound Poisson Process

When the interevent times $\{J_i\}$ are exponentially distributed with parameter λ , the counting process $N(t)$ is a Poisson process, and $Z(t)$ is a compound Poisson process.

- In this case, the sample paths of $Z(t)$ have jumps at random times following a Poisson process with rate λ .
- The jump sizes are i.i.d. according to the distribution of X_i .
- The process has independent increments: for any $0 \leq s < t$, the increment $Z(t) - Z(s)$ is independent of $\{Z(u) : 0 \leq u \leq s\}$.
- The process has stationary increments: the distribution of $Z(t) - Z(s)$ depends only on the difference $t - s$.

4 Conditions for Markov Property of $Z(t)$

In this section I determine the conditions under which the compound renewal process $Z(t)$ is a Markov process.

4.1 Definition of Markov Property

A stochastic process $\{Z(t), t \geq 0\}$ is a Markov process if for any times $0 \leq t_1 < t_2 < \dots < t_n < t$ and any Borel set B :

$$P(Z(t) \in B | Z(t_1), Z(t_2), \dots, Z(t_n)) = P(Z(t) \in B | Z(t_n))$$

This property, known as the Markov property, states that the future behavior of the process depends only on its current state, not on its past history.

4.2 Analysis for General Compound Renewal Processes

For a general compound renewal process, we need to examine whether the conditional distribution of $Z(t)$ given the past depends only on the most recent value $Z(t_n)$.

Let's consider the evolution of $Z(t)$ from time t_n to time t . We have:

$$\begin{aligned}
Z(t) &= \sum_{i=1}^{N(t)} X_i \\
&= \sum_{i=1}^{N(t_n)} X_i + \sum_{i=N(t_n)+1}^{N(t)} X_i \\
&= Z(t_n) + \sum_{i=N(t_n)+1}^{N(t)} X_i
\end{aligned}$$

The increment $\sum_{i=N(t_n)+1}^{N(t)} X_i$ depends on two factors:

1. The number of new events in the interval $(t_n, t]$, given by $N(t) - N(t_n)$
2. The jump sizes X_i for $i = N(t_n) + 1, \dots, N(t)$

While the jump sizes X_i are i.i.d. and independent of the past, the distribution of $N(t) - N(t_n)$ given the past depends on the specific properties of the renewal process $N(t)$.

For a general renewal process, the distribution of $N(t) - N(t_n)$ depends not only on $N(t_n)$ but also on the elapsed time since the last renewal before t_n . This is known as the "age" of the process at time t_n and is defined as:

$$A(t_n) = t_n - T_{N(t_n)}$$

Given only $Z(t_n)$, we don't have information about $A(t_n)$, which means that $Z(t)$ is generally not a Markov process for an arbitrary renewal process.

4.3 Condition for Markov Property: Memoryless Interevent Times

The key condition for $Z(t)$ to be a Markov process is that the interevent times $\{J_i\}_{i=1}^{\infty}$ must have the memoryless property. This is satisfied when:

$$P(J_i > t + s | J_i > s) = P(J_i > t) \quad \text{for all } t, s \geq 0$$

It is well-known in probability theory that the only continuous distribution with the memoryless property is the exponential distribution [6]. Therefore:

The compound renewal process $Z(t) = \sum_{i=1}^{N(t)} X_i$ is a Markov process if and only if the interevent times $\{J_i\}_{i=1}^{\infty}$ are exponentially distributed.

When the interevent times are exponentially distributed with parameter λ , the renewal process $N(t)$ becomes a Poisson process with rate λ , and $Z(t)$ is called a compound Poisson process.

4.4 Proof of Markov Property for Compound Poisson Processes

Here is a formal proof that a compound Poisson process $Z(t)$ is a Markov process.

Suppose the interevent times $\{J_i\}_{i=1}^{\infty}$ are exponentially distributed with parameter λ , so that $N(t)$ is a Poisson process with rate λ .

For any times $0 \leq t_1 < t_2 < \dots < t_n < t$ and any Borel set B , we need to show:

$$P(Z(t) \in B | Z(t_1), Z(t_2), \dots, Z(t_n)) = P(Z(t) \in B | Z(t_n))$$

First, note that for a Poisson process $N(t)$, the increments are independent and stationary. This means that for any times $s < t$, the random variable $N(t) - N(s)$ is independent of $\{N(u) : 0 \leq u \leq s\}$ and follows a Poisson distribution with parameter $\lambda(t - s)$.

Now, we can write:

$$\begin{aligned} Z(t) &= Z(t_n) + [Z(t) - Z(t_n)] \\ &= Z(t_n) + \sum_{i=N(t_n)+1}^{N(t)} X_i \end{aligned}$$

Given $Z(t_1), Z(t_2), \dots, Z(t_n)$, the distribution of $Z(t)$ depends on:

1. The value of $Z(t_n)$
2. The distribution of $\sum_{i=N(t_n)+1}^{N(t)} X_i$

For the second term, we note that:

1. The increment $N(t) - N(t_n)$ is independent of $\{N(u) : 0 \leq u \leq t_n\}$ due to the independent increments property of the Poisson process.
2. The random variables X_i for $i > N(t_n)$ are independent of $X_1, X_2, \dots, X_{N(t_n)}$ by the i.i.d. assumption.

3. Therefore, the sum $\sum_{i=N(t_n)+1}^{N(t)} X_i$ is independent of $Z(t_1), Z(t_2), \dots, Z(t_n)$ given $N(t_n)$.

For a compound Poisson process, the distribution of future increments depends only on the time difference $(t - t_n)$ and not on the history of the process. This is a direct consequence of the memoryless property of the exponential distribution.

Therefore:

$$\begin{aligned} P(Z(t) \in B | Z(t_1), Z(t_2), \dots, Z(t_n)) &= P(Z(t_n) + [Z(t) - Z(t_n)] \in B | Z(t_1), Z(t_2), \dots, Z(t_n)) \\ &= P(Z(t_n) + [Z(t) - Z(t_n)] \in B | Z(t_n)) \\ &= P(Z(t) \in B | Z(t_n)) \end{aligned}$$

This establishes that $Z(t)$ is a Markov process when the interevent times are exponentially distributed.

5 Conditions for Martingale Property of $Z(t)$

In this section we investigate the conditions under which the compound renewal process $Z(t)$ is a martingale.

5.1 Definition of a Martingale

A stochastic process $\{Z(t), t \geq 0\}$ is a martingale with respect to a filtration $\{\mathcal{F}_t, t \geq 0\}$ if the following conditions are satisfied:

1. Adaptedness: $Z(t)$ is $\mathcal{F}_t$ -measurable for all $t \geq 0$
2. Integrability: $\mathbb{E}[|Z(t)|] < \infty$ for all $t \geq 0$
3. Martingale property: $\mathbb{E}[Z(t) | \mathcal{F}_s] = Z(s)$ for all $0 \leq s < t$

The martingale property implies that the best prediction of the future value of the process, given all information up to the present, is simply the present value [9].

5.2 Filtration for Compound Renewal Processes

To analyze the martingale property, we need to specify an appropriate filtration. The natural filtration for the compound renewal process is:

$$\mathcal{F}_t = \sigma(N(u), X_1, X_2, \dots, X_{N(t)} : 0 \leq u \leq t)$$

This filtration represents the information available up to time t , including the history of the counting process and the values of the jumps that have occurred by time t .

5.3 Verification of Adaptedness Condition

First, we verify that $Z(t)$ is adapted to the filtration $\{\mathcal{F}_t, t \geq 0\}$, meaning that $Z(t)$ is $\mathcal{F}_t$ -measurable for all $t \geq 0$.

Since $Z(t) = \sum_{i=1}^{N(t)} X_i$, it is a function of $N(t)$ and $X_1, X_2, \dots, X_{N(t)}$, all of which are included in $\mathcal{F}_t$. Therefore, $Z(t)$ is $\mathcal{F}_t$ -measurable, and the adaptedness condition is satisfied.

5.4 Verification of Integrability Condition

Next, we verify that the integrability condition $\mathbb{E}[|Z(t)|] < \infty$ is satisfied for all $t \geq 0$.

For a compound renewal process, we have:

$$\begin{aligned} \mathbb{E}[|Z(t)|] &= \mathbb{E} \left[\left| \sum_{i=1}^{N(t)} X_i \right| \right] \\ &\leq \mathbb{E} \left[\sum_{i=1}^{N(t)} |X_i| \right] \quad (\text{by triangle inequality}) \end{aligned}$$

Using the law of iterated expectations:

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^{N(t)} |X_i| \right] &= \mathbb{E} \left[\mathbb{E} \left[\sum_{i=1}^{N(t)} |X_i| \mid N(t) \right] \right] \\ &= \mathbb{E}[N(t) \cdot \mathbb{E}[|X_1|]] \\ &= \mathbb{E}[|X_1|] \cdot \mathbb{E}[N(t)] \end{aligned}$$

For a renewal process with finite expected interevent time $\mathbb{E}[J_1] < \infty$, the expected number of renewals $\mathbb{E}[N(t)]$ is finite for any finite t . If additionally $\mathbb{E}[|X_1|] < \infty$, then:

$$\mathbb{E}[|Z(t)|] \leq \mathbb{E}[|X_1|] \cdot \mathbb{E}[N(t)] < \infty$$

Therefore, the integrability condition is satisfied when the jumps have finite first absolute moment ($\mathbb{E}[|X_1|] < \infty$) and the interevent times have finite mean ($\mathbb{E}[J_1] < \infty$).

5.5 Analysis of Martingale Property

Now we determine when the compound renewal process $Z(t)$ satisfies the martingale property:

$$\mathbb{E}[Z(t) \mid \mathcal{F}_s] = Z(s) \quad \text{for all } 0 \leq s < t$$

We can express $Z(t)$ as:

$$\begin{aligned} Z(t) &= \sum_{i=1}^{N(t)} X_i \\ &= \sum_{i=1}^{N(s)} X_i + \sum_{i=N(s)+1}^{N(t)} X_i \\ &= Z(s) + \sum_{i=N(s)+1}^{N(t)} X_i \end{aligned}$$

Therefore:

$$\begin{aligned} \mathbb{E}[Z(t) \mid \mathcal{F}_s] &= \mathbb{E}[Z(s) + \sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s] \\ &= Z(s) + \mathbb{E}[\sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s] \end{aligned}$$

For $Z(t)$ to be a martingale, we need:

$$\mathbb{E}[\sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s] = 0 \quad \text{for all } 0 \leq s < t$$

In the standard formulation of compound renewal processes, the sequence $\{X_i\}_{i=1}^\infty$ is independent of the counting process $N(t)$. Using this independence and the law of iterated expectations:

$$\mathbb{E}\left[\sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s\right] = \mathbb{E}\left[\mathbb{E}\left[\sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s, N(t) - N(s)\right] \mid \mathcal{F}_s\right]$$

Given $\mathcal{F}_s$ and $N(t) - N(s)$, the inner expectation is:

$$\mathbb{E}\left[\sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s, N(t) - N(s)\right] = (N(t) - N(s)) \cdot \mathbb{E}[X_1]$$

since the X_i 's are i.i.d. with mean $\mathbb{E}[X_1]$ and independent of the counting process.

Therefore:

$$\begin{aligned} \mathbb{E}\left[\sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s\right] &= \mathbb{E}[(N(t) - N(s)) \cdot \mathbb{E}[X_1] \mid \mathcal{F}_s] \\ &= \mathbb{E}[X_1] \cdot \mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s] \end{aligned}$$

For this expression to be zero for all $s < t$, we need $\mathbb{E}[X_1] = 0$ or $\mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s] = 0$ for all $s < t$.

The second condition ($\mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s] = 0$) is generally not possible for a non-trivial renewal process, as the expected number of events in a future interval is typically positive and finite. Therefore, we obtain our necessary and sufficient condition:

$$\mathbb{E}[X_1] = 0$$

This means that the expected value of the jumps must be zero for $Z(t)$ to be a martingale.

5.6 Proof Details for Different Types of Renewal Processes

While the condition $\mathbb{E}[X_1] = 0$ is necessary and sufficient for all compound renewal processes, the proof details differ slightly between compound Poisson processes and general compound renewal processes.

5.6.1 Compound Poisson Process

When the interevent times $\{J_i\}$ are exponentially distributed with parameter λ , the counting process $N(t)$ is a Poisson process with rate λ .

For a Poisson process, the increment $N(t) - N(s)$ is independent of $\mathcal{F}_s$ and follows a Poisson distribution with parameter $\lambda(t - s)$. Therefore:

$$\mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s] = \mathbb{E}[N(t) - N(s)] = \lambda(t - s)$$

Substituting this into our expression:

$$\mathbb{E}\left[\sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s\right] = \mathbb{E}[X_1] \cdot \lambda(t - s)$$

For this to be zero for all $s < t$, we need $\mathbb{E}[X_1] = 0$, since $\lambda(t - s) > 0$ for $t > s$.

5.6.2 General Compound Renewal Process

For a general renewal process, the increment $N(t) - N(s)$ can depend on $\mathcal{F}_s$, particularly on the age of the process at time s , which is defined as:

$$A(s) = s - T_{N(s)}$$

Due to this dependence, $\mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s]$ will typically be a function of $A(s)$ and will vary based on the history of the process. However, for any non-trivial renewal process, this conditional expectation is positive and finite for $t > s$ [7].

Therefore, the condition $\mathbb{E}[X_1] = 0$ is still necessary and sufficient for a general compound renewal process to be a martingale, since:

$$\mathbb{E}\left[\sum_{i=N(s)+1}^{N(t)} X_i \mid \mathcal{F}_s\right] = \mathbb{E}[X_1] \cdot \mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s] = 0 \cdot (\text{finite value}) = 0$$

regardless of the specific value of $\mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s]$.

5.7 Summary of Martingale Conditions

Based on the analysis above, we can summarize the condition under which the compound renewal process $Z(t)$ is a martingale:

The compound renewal process $Z(t) = \sum_{i=1}^{N(t)} X_i$ is a martingale with respect to the filtration $\mathcal{F}_t = \sigma(N(u), X_1, X_2, \dots, X_{N(t)} : 0 \leq u \leq t)$ if and only if:

$$\mathbb{E}[X_1] = 0$$

That is, the mean of the jump distribution must be zero.

This condition applies to both compound Poisson processes and general compound renewal processes, though the proof details differ due to the different properties of the underlying counting processes.

5.8 Examples

Example 1: Consider a compound Poisson process with rate $\lambda = 2$ where the jumps X_i follow a normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 1$. Since $\mathbb{E}[X_1] = 0$, this process is a martingale.

Example 2: Consider a compound renewal process where the interevent times follow a gamma distribution with shape parameter $\alpha = 2$ and scale parameter $\beta = 1$, and the jumps follow a symmetric Laplace distribution with location parameter 0 and scale parameter 1. The mean of the jumps is 0, so this process is a martingale.

Example 3: Consider a compound Poisson process with rate $\lambda = 3$ where the jumps X_i take values $+1$ with probability 0.4 and -1 with probability 0.6. The mean of X_i is $0.4 \times 1 + 0.6 \times (-1) = -0.2 \neq 0$, so this process is not a martingale.

6 Applications in Queueing Theory

Compound renewal processes find extensive applications in queueing theory, especially for modeling systems with batch arrivals or bulk services. Here, we explore some key applications based on the theoretical foundations developed in previous sections.

6.1 M/G/1 Queue with Batch Arrivals

A natural application of compound Poisson processes is in modeling queueing systems where customers arrive in batches. In the M/G/1 queueing model:

- Customer batches arrive according to a Poisson process with rate λ
- The batch size X follows a general distribution with mean $\mathbb{E}[X]$
- Service times follow a general distribution with mean $\mathbb{E}[S]$

The traffic intensity is given by $\rho = \lambda \mathbb{E}[X] \mathbb{E}[S]$, and the system is stable when $\rho < 1$.

The famous Pollaczek-Khinchin formula can be extended to find the mean queue length:

$$\mathbb{E}[L_q] = \frac{\lambda^2 \mathbb{E}[S^2] \mathbb{E}[X] + \lambda \mathbb{E}[S] \mathbb{E}[X(X-1)]}{2(1-\rho)}$$

This formula explicitly accounts for the additional variability introduced by the batch arrival structure [1].

References

- [1] Tijms, H. C. (2003). *A First Course in Stochastic Models*. John Wiley & Sons Ltd, Chichester, West Sussex, England.
- [2] Serfozo, R. (2009). *Basics of Applied Stochastic Processes*. Springer Series in Operations Research and Financial Engineering. Springer, New York.
- [3] Billingsley, P. (2008). *Probability and Measure*. John Wiley & Sons.
- [4] Garibaldi, U., & Scalas, E. (2010). *Finitary Probabilistic Methods in Econophysics*. Cambridge University Press.
- [5] Last, G., & Brandt, A. (1995). *Marked Point Processes on the Real Line: The Dynamic Approach*. Springer.
- [6] Karlin, S., & Taylor, H. M. (1975). *A First Course in Stochastic Processes*. Academic Press.
- [7] Çinlar, E. (1975). *Introduction to Stochastic Processes*. Prentice-Hall.

- [8] Feller, W. (1971). *An Introduction to Probability Theory and Its Applications*, Vol. 2. Wiley.
- [9] Revuz, D., & Yor, M. (1999). *Continuous Martingales and Brownian Motion*. Springer.